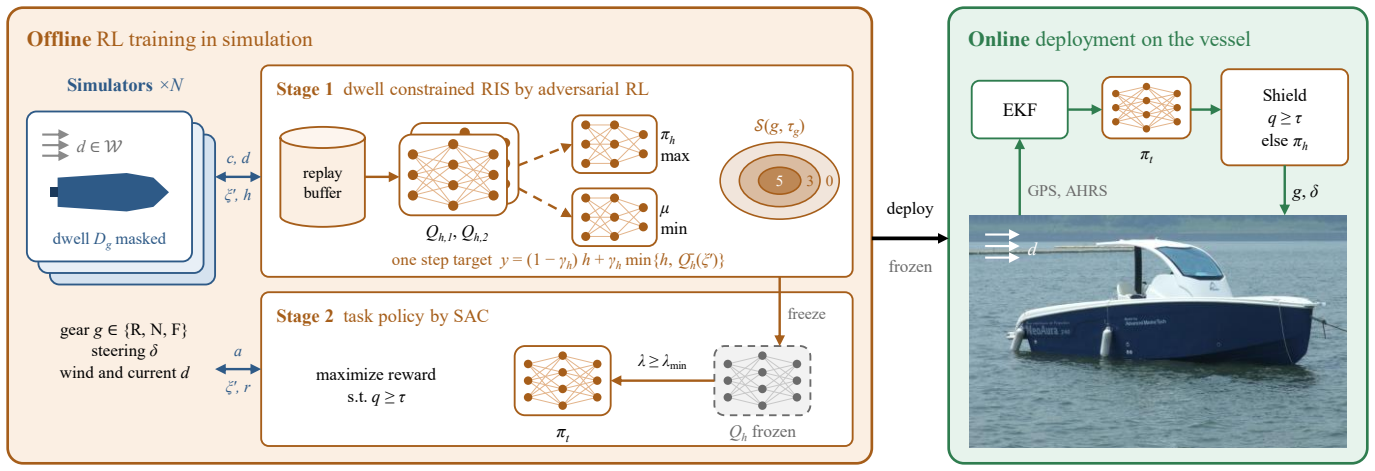

Offline RL training in simulation
Simulators ×N
d ∈ 𝒲
dwell D_g masked
c, d
ξ′, h
Stage 1 dwell constrained RIS by adversarial RL
replay buffer
$Q_{h,1}$, $Q_{h,2}$
π_h max
μ min
$\mathcal{S}(g, \tau_g)$
5
3
0
one step target $y = (1 - \gamma_h)\, h + \gamma_h \min\{h,\ Q_h(\zeta')\}$
gear g ∈ {R, N, F}
steering δ
wind and current d
a
ξ′, r
Stage 2 task policy by SAC
maximize reward
s.t. $q \geq \tau$
freeze
$\lambda \geq \lambda_{\min}$
π_t
Q_h frozen
deploy
frozen
Online deployment on the vessel
EKF
π_t
Shield
$q \geq \tau$
else π_h
GPS, AHRS
g, δ
d